Math 105: Calculus with Algebra
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quiz archive

Gradescope code: B5J8EN

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Math 105 Fall 2026 - Take-Home Quiz 1 Solutions
Due Sunday 9/6/2026 - at 11:55 PM

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(1) Solve the equation $|x-2|=7$ for x.

Solution. Since $|x-2|=7$, either $x-2=7$ or $x-2=-7$. Thus $x=9$ or $x=-5$.

(2) Solve the inequality $|x+3|<4$.

Solution. The inequality $|x+3|<4$ is equivalent to $-4<x+3<4$. Subtracting 3 throughout gives $-7<x<1$.

Thus the solution is $(-7,1)$. On the number line, shade the interval between -7 and 1, with open circles at -7 and 1.

(3) Express

$$1+\frac{1}{\frac{x}{x+2}}$$

as a single fraction.

Solution. We have

$$1+\frac{1}{\frac{x}{x+2}}=1+\frac{x+2}{x}=\frac{x+x+2}{x}=\frac{2x+2}{x}.$$

Thus the single fraction is $\dfrac{2x+2}{x}$.

The original expression is defined only when $x\neq -2$ and $x\neq 0$.

(1) Consider the line

$$x + 2y = 2.$$

(a) To find the slope, solve the equation for y:

$$x + 2y = 2$$

$$2y = -x + 2$$

$$y = -\frac{1}{2}x + 1.$$

Thus the slope is

$$\boxed{-\frac{1}{2}}.$$

(b) A line parallel to this line has the same slope, so its slope is $-\frac{1}{2}$. Since it passes through $(2, -1)$, we can use point-slope form:

$$y - (-1) = -\frac{1}{2}(x - 2).$$

Thus

$$y + 1 = -\frac{1}{2}x + 1,$$

and hence

$$\boxed{y = -\frac{1}{2}x}.$$

(c) A perpendicular line has slope equal to the negative reciprocal of the original slope. Since the original slope is $-\frac{1}{2}$, the perpendicular slope is

$$\boxed{2}.$$

(2) Let

$$f(x) = 3x^2 + 2.$$

We want to compute

$$\frac{f(a+h) - f(a)}{h}.$$

First,

$$f(a+h) = 3(a+h)^2 + 2.$$

Expanding gives

$$f(a+h) = 3(a^2 + 2ah + h^2) + 2 = 3a^2 + 6ah + 3h^2 + 2.$$

Also,

$$f(a) = 3a^2 + 2.$$

Therefore

$$\begin{aligned}\frac{f(a+h) - f(a)}{h} &= \frac{3a^2 + 6ah + 3h^2 + 2 - (3a^2 + 2)}{h} \\ &= \frac{6ah + 3h^2}{h} \\ &= \boxed{6a + 3h}.\end{aligned}$$

(3) Let

$$g(x) = \begin{cases} x^2, & \text{if } x \leq 4, \\ -|x - 4|, & \text{if } x > 4. \end{cases}$$

(a) To compute $g(0)$, notice that $0 \leq 4$, so we use the first part of the definition:

$$g(0) = 0^2 = \boxed{0}.$$

Since $5 > 4$, we use the second part to compute $g(5)$:

$$g(5) = -|5 - 4| = -|1| = \boxed{-1}.$$

(b) For $x \leq 4$, the graph is the parabola

$$y = x^2.$$

Since $x = 4$ is included, there is a closed point at

$$(4, 16).$$

For $x > 4$, we have $x - 4 > 0$, so

$$|x - 4| = x - 4.$$

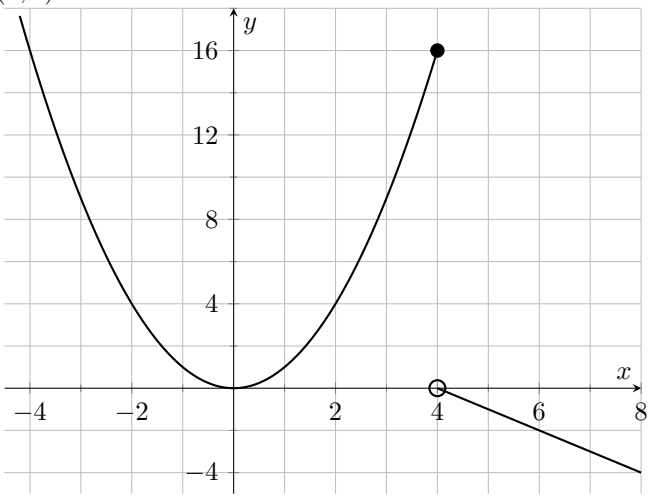
Therefore

$$-|x - 4| = -(x - 4) = 4 - x.$$

Thus the second piece is the line

$$y = 4 - x, \qquad x > 4,$$

with an open point at $(4, 0)$.



The first piece covers all $x \leq 4$ and the second covers all $x > 4$. Therefore the domain is

$$\boxed{(-\infty, \infty)}.$$

The parabola gives every $y \geq 0$, while the line gives every $y < 0$. Together these give every real number. Therefore the range is

$$\boxed{(-\infty, \infty)}.$$

Math 105 Fall 2026 - Take-Home Quiz 3 Solutions

(1) The graph of $y = f(x)$ is shown below, and we want to sketch

$$y = f(x-1) + 2.$$

The expression $f(x-1)$ shifts the graph of f **one unit to the right**. Then adding 2 shifts the graph **two units up**.

Thus every point

$$(x, y)$$

on the original graph moves to

$$(x+1, y+2).$$

In particular,

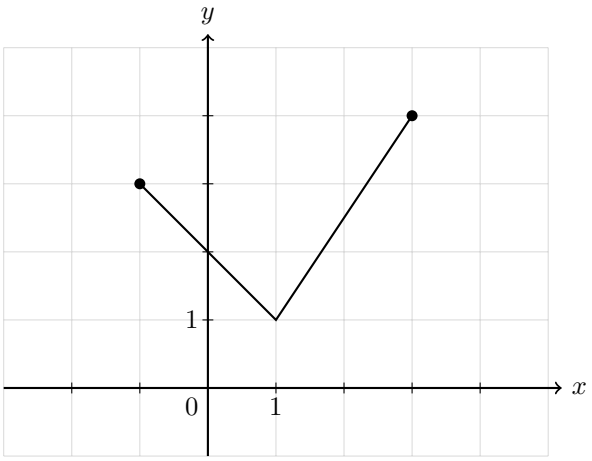
$$(-2, 1) \longmapsto (-1, 3),$$

$$(0, -1) \longmapsto (1, 1),$$

and

$$(2, 2) \longmapsto (3, 4).$$

Therefore the transformed graph is:



Shift right 1 unit and up 2 units.

(2) Let

$$f(x) = \sqrt{x+1} \qquad \text{and} \qquad g(x) = x^2 - 2.$$

We want to compute

$$(f \circ g)(2).$$

Recall that

$$(f \circ g)(2) = f(g(2)).$$

First compute the inside function:

$$g(2) = 2^2 - 2 = 4 - 2 = 2.$$

Then

$$f(g(2)) = f(2) = \sqrt{2+1} = \sqrt{3}.$$

Therefore

$$\boxed{(f \circ g)(2) = \sqrt{3}}.$$

(3) Find

$$\lim_{x\to 3} \frac{x^2-9}{x-3}.$$

If we substitute $x = 3$ immediately, we get

$$\frac{3^2-9}{3-3} = \frac{0}{0},$$

so we need to simplify first.

Factor the numerator:

$$x^2 - 9 = (x-3)(x+3).$$

Therefore, for $x \neq 3$,

$$\frac{x^2-9}{x-3} = \frac{(x-3)(x+3)}{x-3} = x+3.$$

Now we can take the limit:

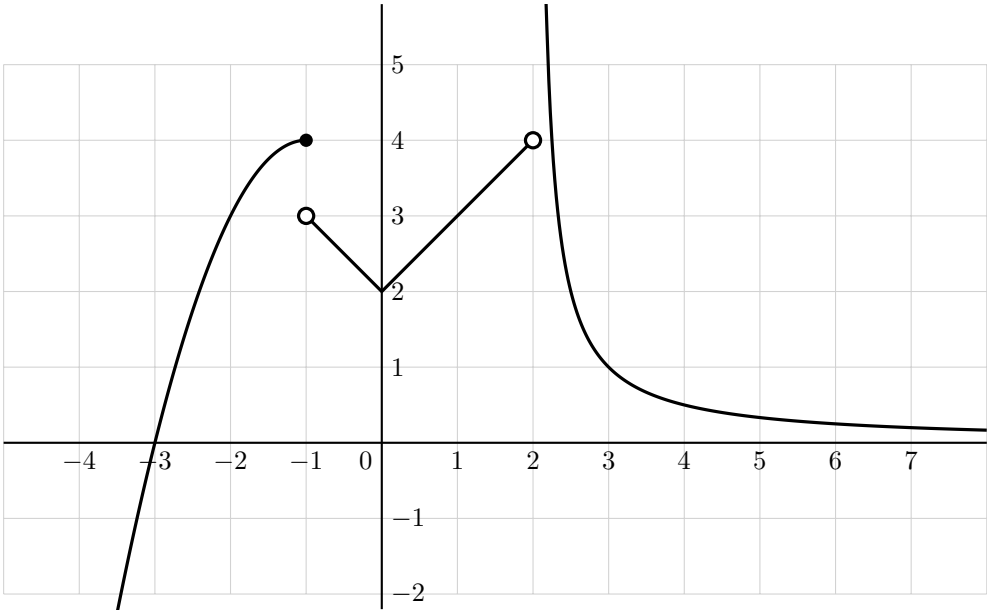
$$\lim_{x\to 3}(x+3) = 3+3 = 6.$$

Thus

$$\boxed{\lim_{x\to 3} \frac{x^2-9}{x-3} = 6}.$$

Math 105 Fall 2026 - Take-Home Quiz 4 Solutions

(1) (5 points) Consider the following graph of $f(x)$.



(a) As x approaches 2 from the right, the graph rises without bound. Therefore

$$\boxed{\lim_{x\to 2^+} f(x) = +\infty.}$$

(b) As x approaches 2 from the left, the graph approaches the open point $(2, 4)$. Therefore

$$\boxed{\lim_{x\to 2^-} f(x) = 4.}$$

(c) The right-hand limit is $+\infty$, while the left-hand limit is 4. Since the one-sided limits do not agree,

$$\boxed{\lim_{x\to 2} f(x) \text{ does not exist.}}$$

(d) As $x \to -1^-$, the graph approaches 4, while as $x \to -1^+$, the graph approaches 3. Since the one-sided limits are different,

$$\boxed{\lim_{x\to -1} f(x) \text{ does not exist.}}$$

(e) Near $x = 0$, the graph lies on the two line segments meeting at $(0, 2)$, so from both sides $f(x)$ approaches 2. Thus

$$\boxed{\lim_{x\to 0} f(x) = 2.}$$

(2) (1 point)

$$\lim_{x\to -3} \frac{x+3}{x^2+x+1}.$$

The denominator is nonzero at $x=-3$, since $(-3)^2+(-3)+1=7$. Therefore direct substitution gives

$$\boxed{0}.$$

(3) (1 point)

$$\lim_{x\to 3} \frac{x-7}{3-x}.$$

As $x\to 3$, the numerator approaches -4. If $x\to 3^-$, then $3-x\to 0^+$, so the quotient approaches $-\infty$. If $x\to 3^+$, then $3-x\to 0^-$, so the quotient approaches $+\infty$. Therefore

$$\boxed{\text{the limit does not exist.}}$$

(4) (1 point)

$$\lim_{x\to -3} \frac{x+3}{x^2-2x-15}.$$

Factor the denominator:

$$x^2-2x-15=(x-5)(x+3).$$

For $x\neq -3$,

$$\frac{x+3}{(x-5)(x+3)}=\frac{1}{x-5}.$$

Therefore

$$\boxed{-\frac{1}{8}}.$$

(5) (2 points)

$$\lim_{x\to -3} \frac{\dfrac{1}{1-x}-\dfrac{1}{4}}{x+3}.$$

Combine the fractions in the numerator:

$$\frac{1}{1-x}-\frac{1}{4}=\frac{4-(1-x)}{4(1-x)}=\frac{x+3}{4(1-x)}.$$

Hence, for $x\neq -3$,

$$\frac{\dfrac{x+3}{4(1-x)}}{x+3}=\frac{1}{4(1-x)}.$$

Now substitute $x=-3$:

$$\boxed{\frac{1}{16}}.$$

Math 105: Midterm 1 Solution Guide

1. Consider the line L_1 given by $3x + 4y = 5$.

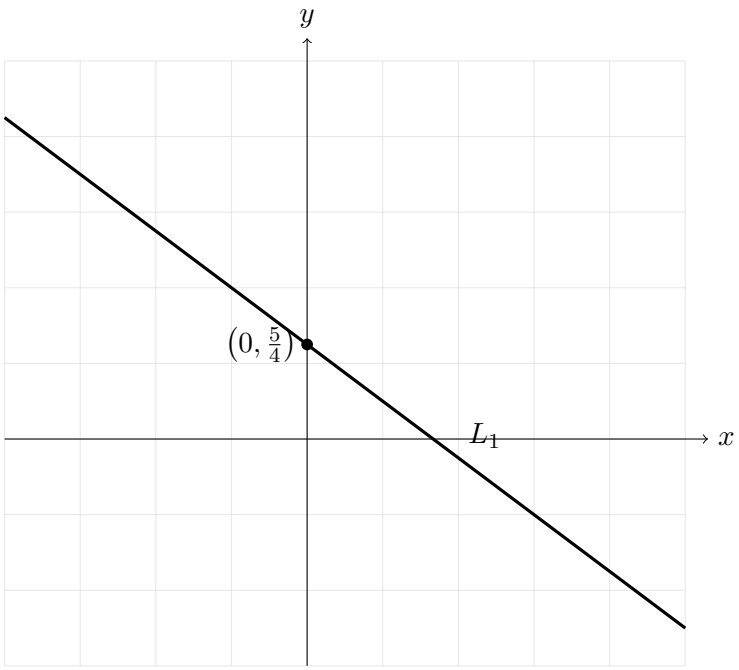
   (a) Solve for y:

   $$3x + 4y = 5 \qquad \Longrightarrow \qquad y = -\frac{3}{4}x + \frac{5}{4}.$$

   Thus L_1 has slope $-\frac{3}{4}$ and y-intercept

   $$\boxed{\left(0, \frac{5}{4}\right)}.$$

   A convenient second point is the x-intercept $\left(\frac{5}{3}, 0\right)$.



   (b) A line perpendicular to L_1 has slope equal to the opposite reciprocal of $-\frac{3}{4}$, namely $\frac{4}{3}$. Using the point $(-5, -1)$,

   $$y + 1 = \frac{4}{3}(x + 5).$$

   Solving for y gives

   $$\boxed{y = \frac{4}{3}x + \frac{17}{3}}.$$

   (c) A line parallel to L_1 must have the same slope. Therefore

   $$\boxed{m = -\frac{3}{4}}.$$

2. Let $f(x) = x^2 - x$ and $g(x) = \sqrt{x + 4}$.

(a) Substitute $2x^3$ for x in f:

$$f(2x^3) = (2x^3)^2 - 2x^3 = \boxed{4x^6 - 2x^3}.$$

(b) Substitute x^2 for x in g:

$$g(x^2) = \boxed{\sqrt{x^2+4}}.$$

(c)

$$(f \circ g)(x) = f(g(x)) = \left(\sqrt{x+4}\right)^2 - \sqrt{x+4}.$$

Therefore

$$\boxed{(f \circ g)(x) = x + 4 - \sqrt{x+4}}.$$

The expression is defined when $x + 4 \geq 0$, so its domain is $[-4, \infty)$.

3. Evaluate each limit.

(a)

$$\lim_{x \to 4} \frac{x^2 - x - 12}{x^2 - 3x - 4}.$$

Direct substitution gives $0/0$, so factor:

$$\frac{(x-4)(x+3)}{(x-4)(x+1)} = \frac{x+3}{x+1} \qquad (x \neq 4).$$

Thus

$$\boxed{\frac{7}{5}}.$$

(b)

$$\lim_{x \to -1} \frac{x^2 - 1}{2 - \sqrt{x+5}}.$$

Again direct substitution gives $0/0$. Rationalize the denominator:

$$\frac{x^2-1}{2-\sqrt{x+5}} \cdot \frac{2+\sqrt{x+5}}{2+\sqrt{x+5}} = \frac{(x-1)(x+1)(2+\sqrt{x+5})}{4-(x+5)}.$$

Since $4 - (x+5) = -(x+1)$, this simplifies to

$$-(x-1)(2+\sqrt{x+5}).$$

Now substitute $x = -1$:

$$-(-2)(2+2) = \boxed{8}.$$

(c)

$$\lim_{x \to 2} \frac{x^2 - 4}{4 - x}.$$

The denominator is nonzero at $x = 2$, so direct substitution works:

$$\frac{4-4}{4-2} = \boxed{0}.$$

(d)

$$\lim_{x\to 2} \frac{x+1}{(x-2)^2}.$$

The numerator approaches $3 > 0$, while $(x-2)^2 \to 0^+$ from both sides. Therefore

$$\boxed{+\infty}.$$

(e)

$$\lim_{x\to 1} \frac{x^2-3x+2}{|x-1|}.$$

Factor the numerator:

$$\frac{(x-1)(x-2)}{|x-1|}.$$

For $x > 1$, $|x-1| = x-1$, so

$$\lim_{x\to 1^+} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^+} (x-2) = -1.$$

For $x < 1$, $|x-1| = -(x-1)$, so

$$\lim_{x\to 1^-} \frac{x^2-3x+2}{|x-1|} = \lim_{x\to 1^-} -(x-2) = 1.$$

The one-sided limits are different, so

$$\boxed{\text{DNE}}.$$

4. Sketch $f(x) = 1 + \dfrac{2}{x-1}$.

The denominator is zero at $x = 1$, so the vertical asymptote is

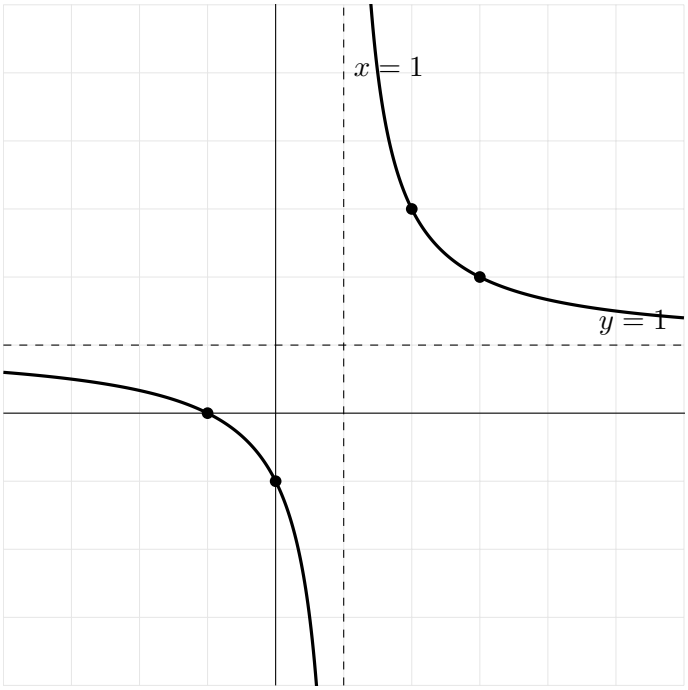
$$\boxed{x = 1}.$$

As x becomes very large in magnitude, $\frac{2}{x-1} \to 0$, so the horizontal asymptote is

$$\boxed{y = 1}.$$

Some convenient points are

$$f(-1) = 0, \qquad f(0) = -1, \qquad f(2) = 3, \qquad f(3) = 2.$$



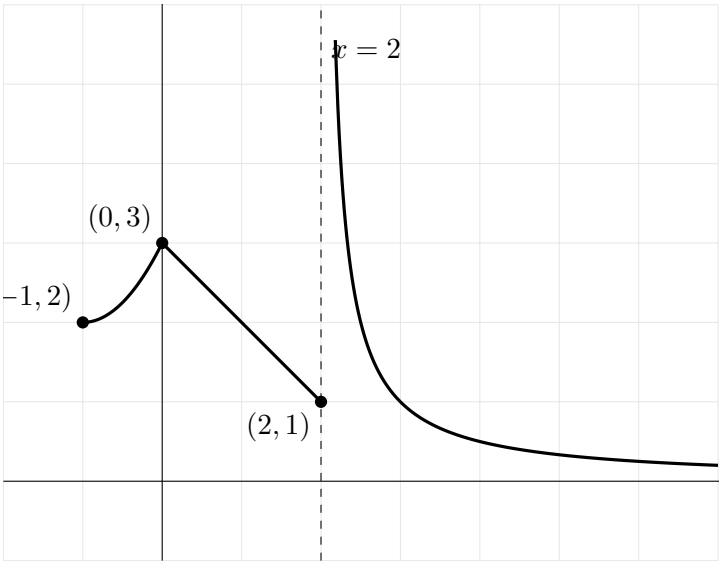
5. Let $f(x) = \dfrac{1}{x}$.

   (a)
   $$\frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h}.$$
   Combine the fractions in the numerator:
   $$\frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} = \frac{-h}{x(x+h)}.$$
   Therefore
   $$\frac{f(x+h) - f(x)}{h} = \boxed{-\frac{1}{x(x+h)}}.$$

   (b) Using the simplified expression from part (a),
   $$\lim_{h \to 0} -\frac{1}{x(x+h)} = \boxed{-\frac{1}{x^2}},$$
   for $x \neq 0$.

6. Consider
   $$f(x) = \begin{cases} \dfrac{1}{x-2}, & x > 2, \\ 3 - |x|, & 0 < x \leq 2, \\ (x+1)^2 + 2, & -1 \leq x \leq 0. \end{cases}$$

   (a) Since $0 < x \leq 2$ in the middle piece, $|x| = x$, so that piece is the line $y = 3 - x$. A sketch is shown below.



(b) Since the middle piece includes $x = 2$,

$$f(2) = 3 - |2| = \boxed{1}.$$

(c) The three pieces cover every input from -1 onward, so

$$\boxed{\text{Domain} = [-1, \infty)}.$$

For $x > 2$, the function $1/(x - 2)$ takes every positive value. Therefore

$$\boxed{\text{Range} = (0, \infty)}.$$

(d) At $x = 0$,

$$\lim_{x \to 0^-} f(x) = 3 \qquad \text{and} \qquad \lim_{x \to 0^+} f(x) = 3.$$

Hence

$$\boxed{\lim_{x \to 0} f(x) = 3}.$$

At $x = 2$,

$$\lim_{x \to 2^-} f(x) = 1, \qquad \lim_{x \to 2^+} f(x) = +\infty.$$

The one-sided limits do not agree, so

$$\boxed{\lim_{x \to 2} f(x) = \text{DNE}}.$$

Math 105 Fall 2026 - Take-Home Quiz 5 Solutions
Due Sunday 10/18/2026 - at 11:55 PM

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(1) Let

$$f(x) = |x|.$$

Using the limit definition,

$$f'(x) = \lim_{h\to 0} \frac{|x+h| - |x|}{h}.$$

Suppose first that $x > 0$. For h sufficiently close to 0, we also have $x + h > 0$, so

$$\frac{|x+h| - |x|}{h} = \frac{(x+h) - x}{h} = 1.$$

Therefore

$$f'(x) = 1 \qquad \text{when } x > 0.$$

Now suppose that $x < 0$. For h sufficiently close to 0, we also have $x + h < 0$, so

$$\frac{|x+h| - |x|}{h} = \frac{-(x+h) - (-x)}{h} = \frac{-h}{h} = -1.$$

Therefore

$$f'(x) = -1 \qquad \text{when } x < 0.$$

At $x = 0$,

$$f'(0) = \lim_{h\to 0} \frac{|h|}{h}.$$

If $h > 0$, then $|h|/h = 1$, while if $h < 0$, then $|h|/h = -1$. The one-sided limits are different, so $f'(0)$ does not exist.

Thus

$$\boxed{f'(x) = \begin{cases} -1, & x < 0, \\ \text{does not exist}, & x = 0, \\ 1, & x > 0. \end{cases}}$$

(2) The matches are

$$\boxed{A \to 2, \qquad B \to 3, \qquad C \to 4, \qquad D \to 1.}$$

- **Function A** is a smooth cubic. Its derivative is the upward-opening parabola in graph 2. Since A is a polynomial, it is differentiable for every real number.
- **Function B** is a smooth parabola with a minimum at $x = 1$. Its derivative is the increasing line in graph 3, which is 0 at $x = 1$. Thus B is differentiable for every real number.
- **Function C** is a V-shaped graph with a corner at $x = 1$. Its slope is $-\frac{1}{2}$ to the left of 1 and $\frac{1}{2}$ to the right, so its derivative is graph 4. It is not differentiable at

$$\boxed{x = 1}.$$

- **Function D** has the same basic V-shape as C, but is steeper. Its slope is -1 to the left of 1 and 1 to the right, so its derivative is graph 1. It is not differentiable at

$$\boxed{x = 1}.$$

Thus the differentiability table is

Function	Where is it not differentiable?
A	nowhere
B	nowhere
C	$x = 1$
D	$x = 1$

(3) Use the power rule (and the sum rule) to compute the derivatives.

(a)

$$f(x) = 4x^5 - 3x^2 + 7x - 9.$$

Differentiate term by term:

$$f'(x) = 20x^4 - 6x + 7.$$

Therefore

$$\boxed{f'(x) = 20x^4 - 6x + 7}.$$

(b)

$$g(x) = \sqrt{x} = x^{1/2}.$$

Using the power rule,

$$g'(x) = \frac{1}{2}x^{-1/2}.$$

Thus, for $x > 0$,

$$\boxed{g'(x) = \frac{1}{2\sqrt{x}}}.$$

(c)

$$h(t) = \frac{1}{t} + t^{1/3} = t^{-1} + t^{1/3}.$$

Using the power rule,

$$h'(t) = -t^{-2} + \frac{1}{3}t^{-2/3}.$$

Therefore, for $t \neq 0$,

$$\boxed{h'(t) = -\frac{1}{t^2} + \frac{1}{3}t^{-2/3}}.$$

Math 105 Fall 2026 - Take-Home Quiz 6 Solutions
Due Sunday 10/25/2026 - at 11:55 PM

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(1) Differentiate the following functions. Simplify your answers when possible.

1a.

$$f(x) = \left(\frac{x^2+1}{x^2-1}\right)^{100}.$$

By the chain rule and quotient rule,

$$f'(x) = 100\left(\frac{x^2+1}{x^2-1}\right)^{99} \frac{2x(x^2-1) - 2x(x^2+1)}{(x^2-1)^2}$$
$$= \boxed{-\frac{400x(x^2+1)^{99}}{(x^2-1)^{101}}}.$$

1b.

$$g(x) = \sqrt{8x^4+\sqrt{x}} = \left(8x^4 + x^{1/2}\right)^{1/2}.$$

Thus

$$g'(x) = \frac{1}{2}\left(8x^4+x^{1/2}\right)^{-1/2}\left(32x^3 + \frac{1}{2\sqrt{x}}\right)$$
$$= \boxed{\frac{32x^3 + \frac{1}{2\sqrt{x}}}{2\sqrt{8x^4+\sqrt{x}}}} \qquad (x>0).$$

(2) Let

$$F(x) = (3x-2)^5(5x+1)^6.$$

Using the product and chain rules,

$$F'(x) = 15(3x-2)^4(5x+1)^6 + 30(3x-2)^5(5x+1)^5$$
$$= 15(3x-2)^4(5x+1)^5(11x-3).$$

Therefore $F'(x) = 0$ when

$$3x-2=0, \qquad 5x+1=0, \qquad 11x-3=0,$$

so

$$\boxed{x = \frac{2}{3}, \quad -\frac{1}{5}, \quad \frac{3}{11}}.$$

(3) Let

$$H(x) = f(g(x)), \qquad K(x) = g(x^2-4).$$

For H, the chain rule gives

$$H'(x) = f'(g(x))g'(x).$$

Since $g(1)=2$, $f'(2)=-4$, and $g'(1)=4$,

$$\boxed{H'(1) = (-4)(4) = -16}.$$

For K,

$$K'(x) = g'(x^2-4)(2x).$$

At $x=2$, we have $x^2-4=0$ and $g'(0)=3$, so

$$\boxed{K'(2) = g'(0)(4) = 3\cdot 4 = 12}.$$